

Views

```
mysql> select * from employee;
+-----+-----+-----+-----+-----+
| employeeid | firstname | lastname | deviceserial | salary |
+-----+-----+-----+-----+-----+
| 1 | John | Smith | ABC123 | 60000 |
| 2 | Jane | Doe | DEF456 | 65000 |
| 3 | Bob | Johnson | GHI789 | 70000 |
| 4 | Sally | Fields | JKL012 | 75000 |
| 5 | Michael | Smith | MNO345 | 80000 |
| 6 | Emily | Jones | PQR456 | 85000 |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> desc employee;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| employeeid | int | YES | | NULL | |
| firstname | varchar(50) | YES | | NULL | |
| lastname | varchar(50) | YES | | NULL | |
| deviceserial | varchar(15) | YES | | NULL | |
| salary | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.02 sec)
```

1.

```
mysql> create view v1 as select employeeid,salary from employee;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> desc v1;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| employeeid | int | YES | | NULL | |
| salary | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2.

```
mysql> insert into v1 values(1001,2000000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from v1;
+-----+-----+
| employeeid | salary |
+-----+-----+
| 1 | 60000 |
| 2 | 65000 |
| 3 | 70000 |
| 4 | 75000 |
| 5 | 80000 |
| 6 | 85000 |
| 101 | 100000 |
| 1001 | 2000000 |
+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> delete from v1 where employeeid=101;
Query OK, 1 row affected (0.01 sec)

mysql> select * from v1;
+-----+-----+
| employeeid | salary |
+-----+-----+
| 1          | 60000  |
| 2          | 65000  |
| 3          | 70000  |
| 4          | 75000  |
| 5          | 80000  |
| 6          | 85000  |
| 1001       | 2000000|
+-----+-----+
7 rows in set (0.00 sec)
```

```
mysql> delete from v1 where employeeid=101;
Query OK, 1 row affected (0.01 sec)

mysql> select * from v1;
+-----+-----+
| employeeid | salary |
+-----+-----+
| 1          | 60000  |
| 2          | 65000  |
| 3          | 70000  |
| 4          | 75000  |
| 5          | 80000  |
| 6          | 85000  |
| 1001       | 2000000|
+-----+-----+
7 rows in set (0.00 sec)
```

Index

1.

```
mysql> SELECT * FROM employee;
+-----+-----+-----+-----+-----+
| employeeid | firstname | lastname | deviceserial | salary |
+-----+-----+-----+-----+-----+
| 1 | John | Smith | ABC123 | 60000 |
| 2 | Jane | Doe | DEF456 | 65000 |
| 3 | Bob | Johnson | GHI789 | 70000 |
| 4 | Sally | Fields | JKL012 | 75000 |
| 5 | Michael | Smith | MNO345 | 80000 |
| 6 | Emily | Jones | PQR456 | 85000 |
| 7 | David | Williams | STU901 | 90000 |
| 8 | Sarah | Johnson | VWX234 | 95000 |
| 9 | James | Brown | YZA567 | 100000 |
| 10 | Emma | Miller | BCD890 | 105000 |
| 11 | William | Davis | EFG123 | 110000 |
| 12 | Olivia | Garcia | HIJ456 | 115000 |
| 13 | Christopher | Rodriguez | KLM789 | 120000 |
| 14 | Isabella | Wilson | NOP012 | 125000 |
| 15 | Matthew | Martinez | QRS345 | 130000 |
| 16 | Sophia | Anderson | TUV678 | 135000 |
| 17 | Daniel | Smith | WXY901 | 140000 |
| 18 | Mia | Thomas | ZAB234 | 145000 |
| 19 | Joseph | Hernandez | CDE567 | 150000 |
| 20 | Abigail | Smith | FGH890 | 155000 |
+-----+-----+-----+-----+-----+
20 rows in set (0.00 sec)
```

```
mysql> select firstname from employee where salary>145000;
```

firstname
Joseph
Abigail

```
2 rows in set (0.00 sec)
```



```
mysql> explain select firstname from employee where salary>145000;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	employee	NULL	ALL	NULL	NULL	NULL	NULL	20	33.33	Using where

```
1 row in set, 1 warning (0.00 sec)
```



```
mysql> create index idx1 on employee(salary);  
Query OK, 0 rows affected (0.07 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```



```
mysql> explain select firstname from employee where salary>145000;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	employee	NULL	range	idx1	idx1	5	NULL	2	100.00	Using index condition

```
1 row in set, 1 warning (0.00 sec)
```

```
mysql> select firstname from employee where salary>145000;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	employee	NULL	ALL	NULL	NULL	NULL	NULL	20	33.33	Using where

```
1 row in set (0.00 sec)
```

```
mysql> explain select firstname from employee where salary>145000;
```

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	employee	NULL	ALL	NULL	NULL	NULL	NULL	20	33.33	Using where

```
1 row in set, 1 warning (0.00 sec)
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mysql> explain select firstname from employee where salary>145000;
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id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	employee	NULL	range	idx1	idx1	5	NULL	2	100.00	Using index condition

```
1 row in set, 1 warning (0.00 sec)
```

[illegible]